Konageshwaran

Homework: 4

Section 2.1:

3.

a) $S = \{TTTT, TTTF, TTFT, TTFF, TFTT, TFTF, TFFT, TFFF, FTTT, FTTF, FTFT, FTFF, FFTT, FFTF, FFFT, FFFF\}$

b) $b = \{TTTT, FFFF\} \Rightarrow P(b) = \frac{2}{16} = \frac{1}{8}$

$P(b) = 1/8$

c) $c = \{TFFF, FTFF, FFTF, FFFT\}$

$P(C) = 4/16 = 1/4$

$P(c) = 1/4$

d) $d = \{FFFF, TFFF, FTFF, FFTF, FFFT\}$

$P(d) = 5/16$

4)

a) conforming = c, downgraded = d, scrap = s

$S = \{$ ccc, ccd, ccs, cdc, cdd, cds, csc, csd, css, dcc, dcd, dcs, ddc, ddd, dds, dsc, dsd, dss, scc, scd, scs, sdc, sdd, sds, ssc, ssd, sss $\}$

123
213
231
312
321

(b) $A = \{ sss, ccc, ddd \}$

(c) $B = \{ cds, dcs, dsc, scd, sdc \}$

(d) $C = \{ ccc, ccd, ccs \}$

(e) $(A \cap C) = \{ ccc \}$

(f) $(A \cup B) = \{ sss, ccc, ddd, cds, dcs, dsc, scd, sdc \}$

(g) $(A \cap C^c) = \{ sss, ddd \}$

(h) $(A^c \cap C) = \{ ccd, ccs \}$

(i) No, A and C share "ccc".

(j) Yes, B and C does not share any common variables.

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(a) $(A \cup B)$? where A = 18-44 age, B = 45-64 age

$P(A \cup B) = 0.19 + 0.30 - P(A \cap B) = 0.19 + 0.30$

$P(A \cup B) = 0.49$

(b) $P(>85) = P(A)$ ⇒ then $P(A^c) = 1 - P(A) = P(A^c) = P(<85)$

$P(A^c) = 1 - 0.04 = 0.96$

$\boxed{P(A) = 0.96}$

(8)

time	Percent
0	71%
1-30	18%
31-90	9%
90<	2%

(a) $P(A)$ = Probability that the bill was paid on time

$\boxed{P(A) = 0.71}$

(b) $P(B)$ = Probability that the bill was paid late.

$P(B) = 1 - P(A)$ because $P(A)$ is on time and $P(A')$ = everything paid late.

$P(B) = 1 - 0.71 = 0.29$

$\boxed{P(B) = 0.29}$

(9)

Status	Percent
flawless	80%
minor	15%
major	5%

(a) $P(A) = P(a) + P(b) - P(a \text{ and } b)$

$P(a)$ = minor, $P(b)$ = major and were mutually Exclusive

$\therefore P(A) = 0.15 + 0.5 = 0.20$

$\boxed{P(A) = 0.20}$

(b) $P(B) = P(a) + P(c) - P(a \cap c)$, where $P(a)$ = minor
$P(c)$ = flawless and they are M.S.

$P(B) = 0.80 + 0.15 = 0.95$

$\boxed{P(B) = 0.95}$

(12)

(a) False

(b) True

(c) True

(14) $P(V) = 0.15$ $P(W) = 0.05$ $P(V \cup W) = 0.17$

(a) $P(V \cap W) = P(V) + P(W) - P(V \cup W)$

$P(V \cap W) = 0.15 + 0.05 - 0.17 = 0.03$

$\boxed{P(V \cap W) = 0.03}$

(b) $P(B) = P(\text{No both}) = 1 - P(V \cup W) = 1 - 0.17 = 0.83$

$\boxed{P(B) = 0.83}$

(c)

$P(C) = P(\text{virus}) - P(V \cap W)$

$P(C) = 0.15 - 0.03 = 0.12$

$\boxed{P(C) = 0.12}$

(16) $n = 600$

(a)

Status	number
cracks	15
discolored	27
good	562

$P(c) = 0.025$

$P(d) = 0.045$

$P(g) = 0.937$

(a)

$P(g) = 1 - P(c \cup d) \Rightarrow P(c \cup d) = 1 - P(g)$

$\Rightarrow P(c \cup d) = 1 - 0.937 = 0.063$

$\boxed{P(c \cup d) = 0.063}$

(b) $P(c \cap d) = P(c) + P(d) - P(c \cup d)$

$P(c \cap d) = 0.025 + 0.045 - 0.063$

$\boxed{P(c \cap d) = 0.007}$

(c) $P(A) = P(c) - P(c \cap d)$

$P(A) = 0.025 - 0.007 = 0.018$

$\boxed{P(A) = 0.018}$